# Multi-Precision Multiplier Microarchitecture

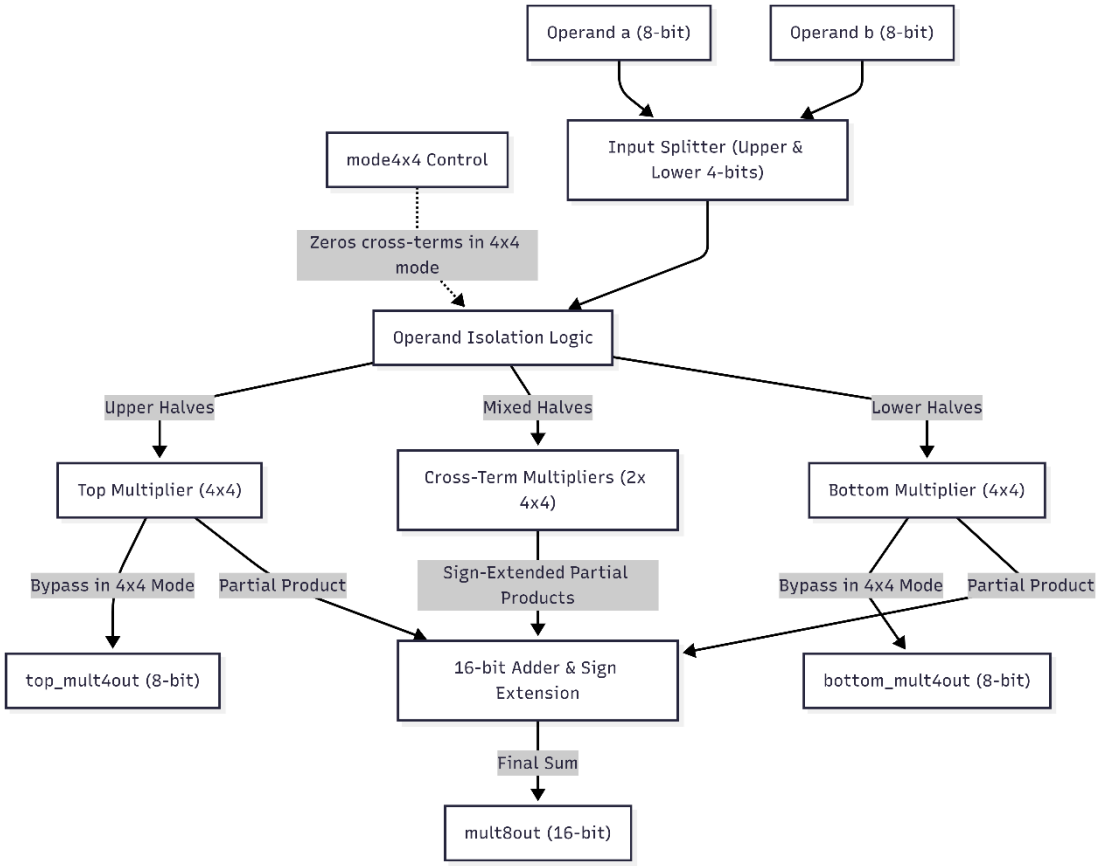


**Figure 1.** Block-level microarchitecture.

## High-Level Overview

The MP_Multiplier is a multi-precision multiplier module capable of executing either a single full 8x8 multiplication or dual independent 4x4 multiplications within a single clock cycle.

## Interface

The MP_Multiplier module interface is designed to support both full 8x8 multiplications and dual independent 4x4 multiplications within a single cycle.

| Port Name | Direction | Width | Description |
|---|---|---|---|
| a | Input | 8-bit bus | Signed input operand. |
| b | Input | 8-bit bus | Signed input operand. |
| mode4x4 | Input | 1 bit | Control signal to set the operation mode. 0 selects a single 8x8 multiplication, and 1 selects dual independent 4x4 multiplications. |
| a_signed | Input | 1 bit | Sign control flag for operand a during 8x8 operations. |
| b_signed | Input | 1 bit | Sign control flag for operand b during 8x8 operations. |
| mult8out | Output | 16-bit bus | Signed 16-bit result of the 8x8 multiplication. |
| bottom_mult4out | Output | 8-bit bus | Signed 8-bit result of the lower 4x4 multiplication. |
| top_mult4out | Output | 8-bit bus | Signed 8-bit result of the upper 4x4 multiplication. |

## Datapath & Operating Modes

The multiplier splits the 8-bit inputs into 4-bit upper and lower halves, routing them to four underlying 4x4 multiplier blocks. The datapath adjusts dynamically based on the mode4x4 signal.

- **INT8 Mode (mode4x4 = 0):** The module functions as a unified 8x8 multiplier. The datapath utilizes all four 4x4 multipliers to compute the partial products.
- **INT4 Mode (mode4x4 = 1):** The module functions as two independent 4x4 multipliers, computing the top and bottom halves simultaneously.
- **Power Optimization (Operand Isolation):** To conserve power during INT4 operations, the module employs an operand isolation strategy. When mode4x4 is driven high, the inputs to the two cross-term multipliers are forced to zero (4'b0000). This effectively freezes the logic inside the unused cross-term multipliers, preventing unnecessary switching activity and saving dynamic power.

## Routing & Final Accumulation

Because the internal 4x4 sub-blocks handle the actual signed math natively, the MP_Multiplier simply routes the correct 4-bit halves and dynamically assigns the sign flags to each block based on the current mode of operation.

- **INT4 Routing (Dual 4x4):** In this mode, the top and bottom multipliers compute their results independently. Their outputs are tapped directly from the first stage, completely bypassing the final addition logic. The top multiplier outputs directly to top_mult4out, and the bottom multiplier outputs directly to bottom_mult4out.
- **INT8 Final Accumulation (Full 8x8):** In this mode, all four multiplier blocks generate partial products. To construct the final 16-bit output, the hardware shifts and adds these partial products together in the main 16-bit adder. The adder outputs to mult8out.
- **INT8 Sign Extension:** Because the cross-product multipliers can generate negative results during an 8x8 operation, their 9-bit outputs must be sign-extended before addition. The datapath extracts the 9th bit from both cross terms. It then explicitly replicates this bit four times to sign-extend the cross products up to 16 bits before feeding them into the final adder.